Aerospace Tech



Project Name: Space Debris Detection System



The Problem: Maya is learning about satellites and discovers that many old satellites and broken space parts are floating around Earth. 🛰️ She learns that these pieces can come close to working satellites and damage them. She thinks a satellite should be able to know when a dangerous object is nearby. Maya decides to build a small system that can detect a nearby object and measure its distance.

🎯 **What You Will Design:** The student will build a space debris detection system that detects objects coming close to a satellite model. The system will show the distance on an OLED display and provide alerts using an LED and buzzer when debris comes too close.



📺 **What is an OLED display?**

An OLED Display is a small screen that shows text, numbers, and messages. It helps users see information clearly from the project.

Example: ⌚ Smartwatches use OLED displays to show time, notifications, and health information on a small screen.



Components Required



Arduino Uno



Ultrasonic Sensor



OLED Display



LED



Buzzer

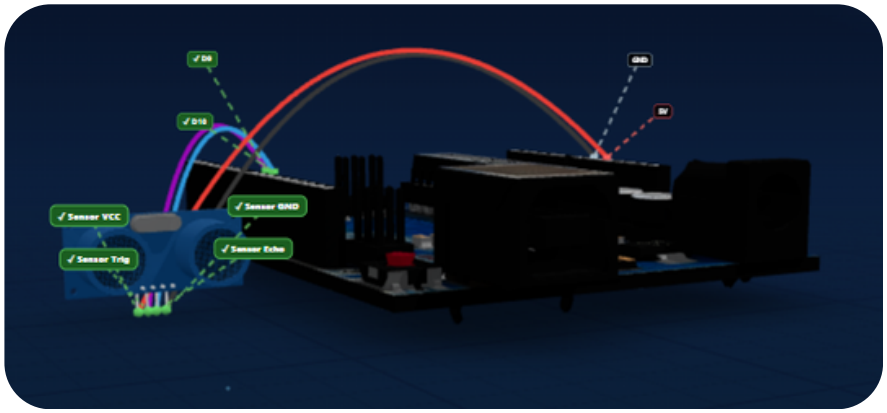


Jumper Wires

Circuit Designing

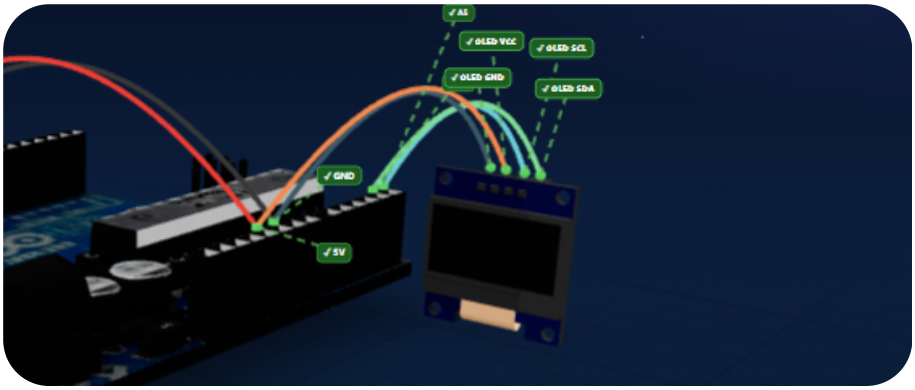
Step 1: Connect the ultrasonic sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
Trig Pin	Digital Pin 9
Echo Pin	Digital Pin 10



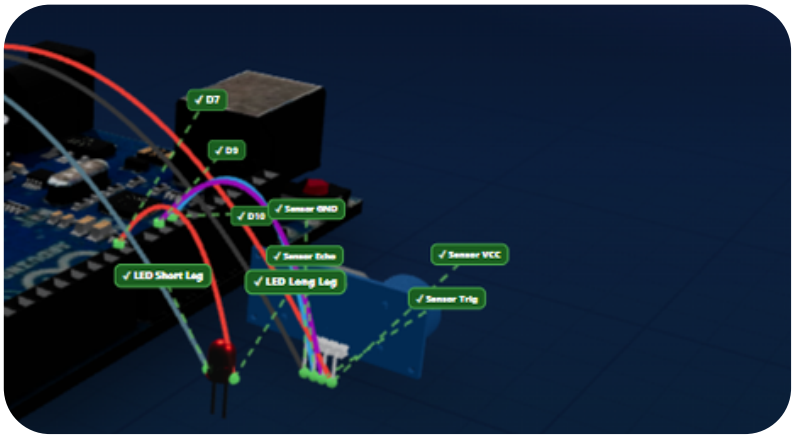
2. Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
SDA Pin	Analog A4 Pin
SCL Pin	Analog A5 Pin



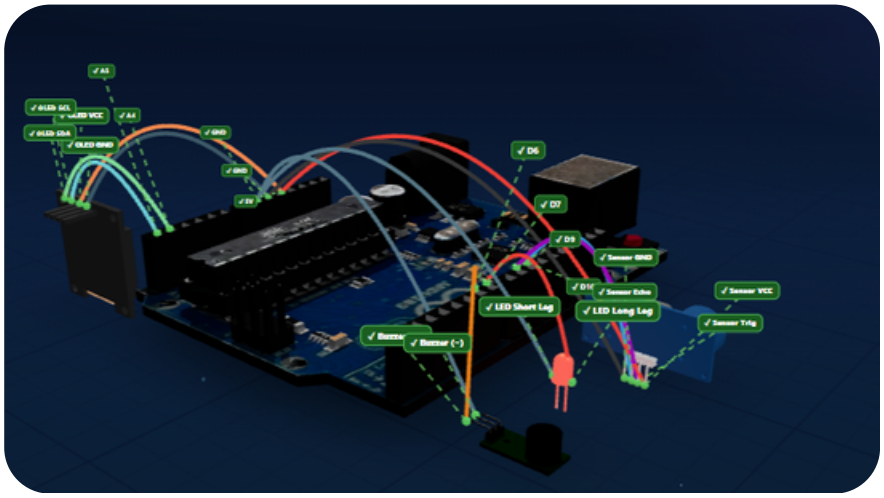
3. Connect the LED to the Arduino Uno.

LED Pins	Arduino Pins
Long Leg	Digital Pin 7
Short Leg	GND Pin



4. Connect the buzzer to the Arduino Uno.

Buzzer Pin	Arduino Pins
Positive Pin	Digital Pin 6
GND Pin	GND Pin



Project Coding

1. Open the Arduino Uno software, add the OLED libraries, and set the ultrasonic sensor pins.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define TRIG_PIN 9
#define ECHO_PIN 10
#define LED_PIN 7
#define BUZZER_PIN 6

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

long duration;
int distance;
```

2. Make code to set up the input and output pins

```
void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  if(!display.begin(SSD1306_I2C_ADDRESS, 0x3C)) {
    for(;;);
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
}
```

3. Make code to calculate the sensor's distance.

```
38  void loop() {
39    digitalWrite(TRIG_PIN, LOW);
40    delayMicroseconds(2);
41    digitalWrite(TRIG_PIN, HIGH);
42    delayMicroseconds(10);
43    digitalWrite(TRIG_PIN, LOW);
44    duration = pulseIn(ECHO_PIN, HIGH);
45    distance = duration * 0.034 / 2;
```

4. Make code to display the distance values on the OLED display.

```
47    display.clearDisplay();
48    display.setCursor(0, 0);
49    display.println("SPACE MONITOR");
50    display.print("Distance: ");
51    display.print(distance);
52    display.println(" cm");
```

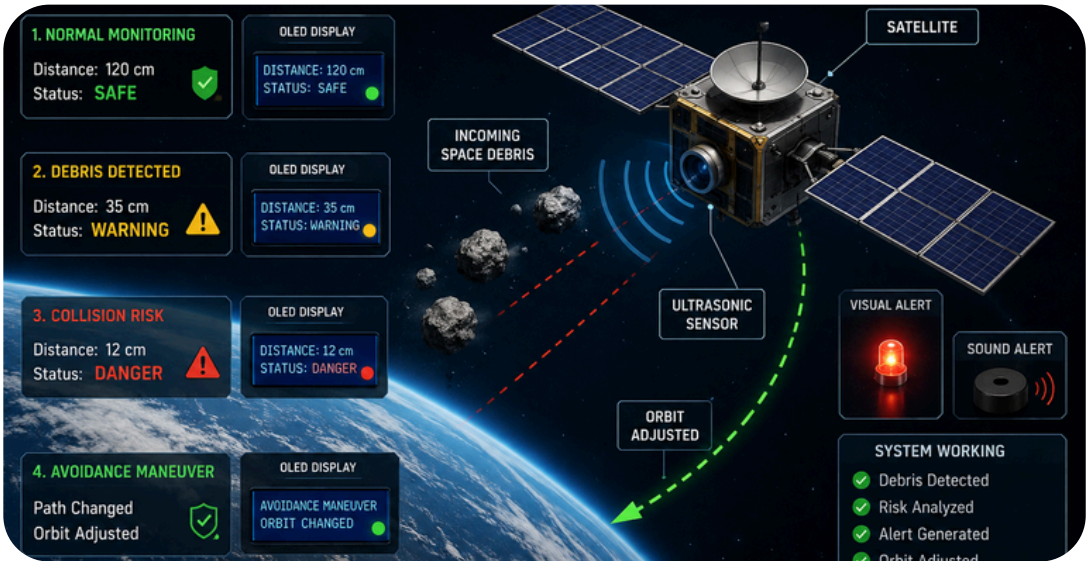
5. Make code to display the 'safe' message on the OLED display.

```
54    if (distance > 20) {
55      display.println("Status: SAFE");
56      digitalWrite(LED_PIN, LOW);
57      digitalWrite(BUZZER_PIN, LOW);
```

6. Make code to display the warning and collision risk messages on the OLED display.

```
59    } else if (distance > 10) {
60      display.println("Status: WARNING");
61      digitalWrite(LED_PIN, LOW);
62      digitalWrite(BUZZER_PIN, LOW);
63    } else {
64      display.println("COLLISION RISK!");
65      digitalWrite(LED_PIN, HIGH);
66      digitalWrite(BUZZER_PIN, HIGH);
67    }
68    display.display();
69    delay(500);
70  }
```

Project Output





🌾 Where Is This Used in Real Life?

Space agencies such as NASA and other satellite organizations continuously monitor space debris around Earth. They track nearby objects to protect satellites and space stations from collisions.

Learning Outcomes

- The students will be able to measure the distance of nearby objects using an ultrasonic sensor.
- The students will be able to code to display live sensor data and warning messages on an OLED screen.
- The students will be able to explain how satellites detect and monitor nearby space debris.



Section 1: Multiple Choice Questions

1. What problem does Maya want to solve?

a) Satellites are changing colour
b) Space debris can come close to working satellites
c) Satellites cannot take pictures
d) Satellites need bigger screens

2. Which sensor is used to detect a nearby object in the project?

a) Soil Sensor
b) RFID Sensor
c) Ultrasonic Sensor
d) Touch Sensor

3. What happens when the debris comes very close?

a) The LED and buzzer give an alert
b) The Arduino turns off
c) The OLED screen disappears
d) The sensor stops measuring

Section 2: True and False Statements

1. Space debris can collide with working satellites and damage them.....

2. The OLED display shows the distance and alert status.....

3. The buzzer is used to measure the distance of space debris.....